

MATH 345 Skeleton Notes

Unit 1: Vectors, matrices, and linear maps

Section 1.4: Matrix multiplication and map composition

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Matrix product by columns

Definition: Matrix-matrix product Let A be an $m \times n$ matrix, let B be an $n \times k$ matrix, and write

$$B = [\mathbf{b}_1 \ \mathbf{b}_2 \ \cdots \ \mathbf{b}_k]$$

where $\mathbf{b}_j$ is column j of B . The product AB is the $m \times k$ matrix

$$AB = [A\mathbf{b}_1 \ A\mathbf{b}_2 \ \cdots \ A\mathbf{b}_k].$$

Activity: Computing a matrix product using columns

Let

$$A = \begin{bmatrix} 1 & -3 & 5 \\ 3 & 1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 1 \\ -2 & -6 \\ -1 & 0 \end{bmatrix}.$$

Compute AB .

The columns of B are $\mathbf{b}_1 = (5, -2, -1)$ and $\mathbf{b}_2 = (1, -6, 0)$. Thus

$$AB = [A\mathbf{b}_1 \ A\mathbf{b}_2] = \begin{bmatrix} 6 & 19 \\ 9 & -3 \end{bmatrix}.$$

Entry formula by row-column dot products

Theorem If A is $m \times n$ and B is $n \times k$, then the (i, j) -entry of AB is the dot product of row i of A with column j of B :

$$(AB)_{ij} = \sum_{\ell=1}^n a_{i\ell} b_{\ell j}.$$

Activity: Computing a matrix entry

If

$$A = \begin{bmatrix} -3 & -4 & 1 \\ 2 & 4 & 0 \\ 1 & -4 & -5 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 2 & -4 \\ -4 & -3 & -1 \\ 4 & 3 & 1 \end{bmatrix},$$

what is the $(2, 3)$ -entry of AB ?

Use row 2 of A and column 3 of B :

$$(2, 4, 0) \cdot (-4, -1, 1) = 2(-4) + 4(-1) + 0(1) = -12.$$

Theorem: Properties of matrix multiplication Assume that A, B, C are matrices of appropriate sizes and r is a scalar. Then

1. $A(BC) = (AB)C$.
2. $(A + B)C = AC + BC$.
3. $C(A + B) = CA + CB$.
4. $A(rB) = r(AB) = (rA)B$.
5. $(AB)^T = B^T A^T$.

Matrix multiplication as composition

If $T_A(\mathbf{x}) = A\mathbf{x}$ and $T_B(\mathbf{x}) = B\mathbf{x}$, then applying B first and then A gives

$$T_A(T_B(\mathbf{x})) = A(B\mathbf{x}) = (AB)\mathbf{x}.$$

Matrix multiplication represents composition of matrix maps.

Warning: Order matters Let

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Then

$$RS = \begin{bmatrix} 0 & -1 \\ 2 & 0 \end{bmatrix}, \quad SR = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix}.$$

Rotating after a horizontal stretch is not the same as stretching after a rotation.

Activity: Composition shape check

Suppose A is 4×3 and B is 3×2 . What is the shape of AB , and what is the domain and codomain of the composed map $\mathbf{x} \mapsto A(B\mathbf{x})$?

The product AB is 4×2 . The composed map takes inputs from $\mathbb{R}^2$ and outputs vectors in $\mathbb{R}^4$.

Activity: Applying one matrix to many points

Put the vertices of a shape in the columns of a matrix X . Then $Y = AX$ contains the transformed vertices.

```
import numpy as np

np.set_printoptions(precision=3, suppress=True)

# Columns are vertices of the unit square.
# The final column repeats the first vertex to close the shape.
X = np.array([
    [0, 1, 1, 0, 0],
    [0, 0, 1, 1, 0],
], dtype=float)

A = np.array([
    [1, 1],
    [0, 1],
], dtype=float)

Y = A @ X
Y
```

Output:

```
array([[0., 1., 2., 1., 0.],
       [0., 0., 1., 1., 0.]])
```

1. What are the columns of X ?
2. What are the columns of Y ?
3. Which transformation from the gallery is this?
4. Why does the bottom edge stay fixed?
5. Why does the top edge shift right?

The columns of X are the vertices of the unit square:

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The final column repeats the first vertex to close the shape. The columns of Y are their images under the map $\mathbf{x} \mapsto A\mathbf{x}$:

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

This matrix is a horizontal shear. It sends (x, y) to $(x + y, y)$. Points with $y = 0$ stay fixed, and points with $y = 1$ shift right by 1.

Warning In this geometric example, the columns of X are points, so $Y = AX$ applies the same matrix to every point. In many data tables, observations are stored as rows. For example, a document matrix may have one document per row. Always check what rows and columns represent before interpreting a product.

Note: Later square-grid visualizations The matrix computation $Y = AX$ applies one matrix to every column of X . For a 2×2 matrix, the images of the coordinate directions determine the whole grid. Later, in Corners are not enough, the same columns-of-points convention

is used for rules that are not matrix maps. Then four corners may not tell the whole story. A grid of input points gives more information.

Many token queries and keys

Example: Many token queries at once Suppose a sequence has L tokens. Store the token vectors as rows of a matrix

$$X = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_L^T \end{bmatrix} \in \mathbb{R}^{L \times d}.$$

A transformer layer forms three views of the same token vectors:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V.$$

The matrices W_Q, W_K, W_V are weight matrices. Their entries are adjusted during training. For now, treat them as ordinary matrices that assign each token three roles.

For token i , the vector $\mathbf{q}_i$ is its query: what token i uses to decide which tokens matter for its update.

For token j , the vector $\mathbf{k}_j$ is its key: what token j uses to be compared against a query.

For token j , the vector $\mathbf{v}_j$ is its value: what token j can contribute after weights are chosen.

Write

$$Q = \begin{bmatrix} \mathbf{q}_1^T \\ \mathbf{q}_2^T \\ \vdots \\ \mathbf{q}_L^T \end{bmatrix}, \quad K = \begin{bmatrix} \mathbf{k}_1^T \\ \mathbf{k}_2^T \\ \vdots \\ \mathbf{k}_L^T \end{bmatrix}.$$

Then

$$K^T = [\mathbf{k}_1 \ \mathbf{k}_2 \ \cdots \ \mathbf{k}_L] .$$

So

$$QK^T = \begin{bmatrix} \mathbf{q}_1^T \\ \mathbf{q}_2^T \\ \vdots \\ \mathbf{q}_L^T \end{bmatrix} [\mathbf{k}_1 \ \mathbf{k}_2 \ \cdots \ \mathbf{k}_L] = \begin{bmatrix} \mathbf{q}_1 \cdot \mathbf{k}_1 & \mathbf{q}_1 \cdot \mathbf{k}_2 & \cdots & \mathbf{q}_1 \cdot \mathbf{k}_L \\ \mathbf{q}_2 \cdot \mathbf{k}_1 & \mathbf{q}_2 \cdot \mathbf{k}_2 & \cdots & \mathbf{q}_2 \cdot \mathbf{k}_L \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{q}_L \cdot \mathbf{k}_1 & \mathbf{q}_L \cdot \mathbf{k}_2 & \cdots & \mathbf{q}_L \cdot \mathbf{k}_L \end{bmatrix} .$$

Thus the score matrix

$$S = QK^T$$

has entries

$$S_{ij} = \mathbf{q}_i \cdot \mathbf{k}_j .$$

Row i of S contains the scores used when updating token i .

Activity: Many-token shape and entry check

Suppose

$$X \in \mathbb{R}^{5 \times 4}$$

and

$$W_Q, W_K, W_V \in \mathbb{R}^{4 \times 3}.$$

Let

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V, \quad S = QK^T.$$

1. What are the shapes of Q , K , and V ?
2. What is the shape of K^T ?
3. What is the shape of $S = QK^T$?
4. Suppose

$$\mathbf{q}_2 = \begin{bmatrix} q_{21} \\ q_{22} \\ q_{23} \end{bmatrix}, \quad \mathbf{k}_4 = \begin{bmatrix} k_{41} \\ k_{42} \\ k_{43} \end{bmatrix}.$$

Expand the entry S_{24} .

5. What does S_{24} compare?
6. What does row 2 of S contain?

Since

$$X \in \mathbb{R}^{5 \times 4} \quad \text{and} \quad W_Q, W_K, W_V \in \mathbb{R}^{4 \times 3},$$

the matrices Q , K , and V all have shape

$$5 \times 3.$$

Thus

$$K^T \in \mathbb{R}^{3 \times 5}.$$

Therefore

$$S = QK^T$$

has shape

$$5 \times 5.$$

The $(2, 4)$ -entry comes from row 2 of Q and column 4 of K^T . But column 4 of K^T is the transpose of row 4 of K . Hence

$$S_{24} = \begin{bmatrix} q_{21} & q_{22} & q_{23} \end{bmatrix} \begin{bmatrix} k_{41} \\ k_{42} \\ k_{43} \end{bmatrix} = q_{21}k_{41} + q_{22}k_{42} + q_{23}k_{43}.$$

So

$$S_{24} = \mathbf{q}_2 \cdot \mathbf{k}_4.$$

This entry compares the query for token 2 with the key for token 4. Row 2 of S contains the scores comparing token 2's query with all five token keys.

A later weighting step converts each row of S into weights that add to 1. Those weights are then used to average the value vectors.

For this section, an

attention calculation

has three steps:

dot-product scores $\longrightarrow$ weights $\longrightarrow$ weighted averages.

The score S_{ij} compares token i 's query with token j 's key. A larger score means token j is more relevant when updating token i .

The scores are not yet the output. For each token i , a weighting rule converts row i of S into numbers

$$\alpha_{i1}, \alpha_{i2}, \dots, \alpha_{iL}$$

that are nonnegative and add to 1. These are the

attention weights

for token i . They say how much token i uses each value vector.

Let

$$A_{\text{att}} = [\alpha_{ij}]$$

be the attention-weight matrix. If the value vectors are stored as rows,

$$V = \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_L^T \end{bmatrix},$$

then the attention output is

$$H = A_{\text{att}} V.$$

Row i of H is

$$\mathbf{h}_i^T = \alpha_{i1} \mathbf{v}_1^T + \alpha_{i2} \mathbf{v}_2^T + \dots + \alpha_{iL} \mathbf{v}_L^T.$$

Equivalently,

$$\mathbf{h}_i = \alpha_{i1} \mathbf{v}_1 + \alpha_{i2} \mathbf{v}_2 + \dots + \alpha_{iL} \mathbf{v}_L.$$

So each updated token vector is a weighted average of value vectors.

Activity: Many-token attention output

Suppose

$$X \in \mathbb{R}^{4 \times 3}, \quad W_Q, W_K, W_V \in \mathbb{R}^{3 \times 2}.$$

Define

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V, \quad S = QK^T.$$

A weighting step converts S into an attention-weight matrix

$$A_{\text{att}} \in \mathbb{R}^{4 \times 4}.$$

The attention output is

$$H = A_{\text{att}}V.$$

1. What are the shapes of Q , K , and V ?
2. What is the shape of $S = QK^T$?
3. What does S_{ij} compare?
4. What does row i of A_{att} tell you?
5. What is the shape of H ?
6. Why is each row of H a weighted average of value vectors?
7. Now suppose row 2 of A_{att} is

$$\alpha_2^T = [0 \quad 1/2 \quad 1/4 \quad 1/4]$$

and

$$V = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 4 \\ 4 & 4 \end{bmatrix}.$$

Since

$$X \in \mathbb{R}^{4 \times 3} \quad \text{and} \quad W_Q, W_K, W_V \in \mathbb{R}^{3 \times 2},$$

the matrices Q , K , and V are all 4×2 .

The score matrix is

$$S = QK^T.$$

Since Q is 4×2 and K^T is 2×4 , the matrix S is 4×4 . The entry S_{ij} compares the query for token i with the key for token j :

$$S_{ij} = \mathbf{q}_i \cdot \mathbf{k}_j.$$

Row i of A_{att} gives the weights token i uses when averaging the value vectors. Since A_{att} is 4×4 and V is 4×2 ,

$$H = A_{\text{att}}V$$

has shape 4×2 .

Row i of H is a weighted average because

$$\mathbf{h}_i = \alpha_{i1} \mathbf{v}_1 + \alpha_{i2} \mathbf{v}_2 + \alpha_{i3} \mathbf{v}_3 + \alpha_{i4} \mathbf{v}_4,$$

where the weights in row i of A_{att} are nonnegative and add to 1.

For row 2,

$$\mathbf{h}_2 = 0 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 2 \\ 0 \end{bmatrix} + \frac{1}{4} \begin{bmatrix} 0 \\ 4 \end{bmatrix} + \frac{1}{4} \begin{bmatrix} 4 \\ 4 \end{bmatrix}.$$

Thus

$$\mathbf{h}_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

The second token ignores value vector 1, uses value vector 2 with weight 1/2, and uses value vectors 3 and 4 with weight 1/4 each.

Warning In real transformer attention, the weighting rule usually uses scaled and masked scores before forming weights. For Unit 1, the important facts are: rows of A_{att} are weights, the weights add to 1, and $A_{\text{att}}V$ forms weighted averages of the rows of V .

Activity: The same shape check in code

The following code performs a many-token shape check similar to the previous activity.

```
import numpy as np

X_tokens = np.ones((5, 4))
WQ = np.ones((4, 3))
WK = np.ones((4, 3))
WV = np.ones((4, 3))

Q = X_tokens @ WQ
K_tokens = X_tokens @ WK
V_tokens = X_tokens @ WV
scores_many = Q @ K_tokens.T

Q.shape, K_tokens.shape, V_tokens.shape, scores_many.shape
```

Output:

```
((5, 3), (5, 3), (5, 3), (5, 5))
```

1. Which line creates all query-key scores at once?
2. Why is `scores_many.shape` equal to `(5, 5)`?

The line `scores_many = Q @ K_tokens.T` creates all query-key scores at once. Since `Q` has shape `(5, 3)` and `K_tokens.T` has shape `(3, 5)`, the score matrix has shape `(5, 5)`.